Namespace Teradyne.FA.DIA.DAS.Adaptive

Classes

AdaptiveCommand

Builds and serializes AdaptiveRequests to JSON for transmission.

AdaptiveCommunication

This class manages RabbitMQ communication to send adaptive request to the tester host.

AdaptiveRequest

Represents an adaptive request sent to the ATE system, including test step actions, enable word changes, and target information.

EnableWordAction

Represents an action applied to a set of enable words for adaptive behavior.

Limits

Defines numerical limits and metadata associated with a test step or limit configuration.

TestStepAction

Represents an action that can be applied to one or more test steps in an adaptive test scenario.

Enums

AdaptiveRequestTarget

Enum specifying the target component for an adaptive request.

EnableWordActionType

Specifies the type of action to apply on enable words in adaptive control.

TestStepActionType

Defines the types of actions that can be taken on test steps in an adaptive scenario.

Class AdaptiveCommand

Namespace: Teradyne.FA.DIA.DAS.Adaptive

Assembly: Adaptive.dll

Builds and serializes AdaptiveRequests to JSON for transmission.

```
public class AdaptiveCommand
```

Inheritance

object ← AdaptiveCommand

Inherited Members

object.Equals(object), object.Equals(object, object), object.GetHashCode(), object.GetType(), object.MemberwiseClone(), object.ReferenceEquals(object, object), object.ToString()

Constructors

AdaptiveCommand(string, List<EnableWordAction>, List<TestStepAction>)

Initializes a new AdaptiveCommand object.

```
public AdaptiveCommand(string DasId, List<EnableWordAction> enableWords = null,
List<TestStepAction> steps = null)
```

Parameters

`DasId` string

DAS identifier string.

`enableWords` List<EnableWordAction>

Optional list of EnableWordAction objects.

`steps` List<TestStepAction>

Optional list of TestStepAction objects.

Methods

addAction(EnableWordAction)

Adds an EnableWordAction to the request.

```
public bool addAction(EnableWordAction action)
```

Parameters

`action` EnableWordAction

Returns

bool

addAction(TestStepAction)

Adds a TestStepAction to the request.

```
public bool addAction(TestStepAction action)
```

Parameters

`action` TestStepAction

Returns

bool

createClearAction()

Creates a TestStepAction that clears all existing configurations.

```
public TestStepAction createClearAction()
```

Returns

TestStepAction

createEnableWordAction(EnableWordActionType, List<string>)

Creates a new EnableWordAction.

```
public EnableWordAction createEnableWordAction(EnableWordActionType actionType,
List<string> enableWords)
```

Parameters

`actionType` EnableWordActionType

`enableWords` List<string>

Returns

EnableWordAction

createLimitsAction(TestStepActionType, List<int>, Limits)

Creates a TestStepAction with limits using step numbers.

```
public TestStepAction createLimitsAction(TestStepActionType actionType, List<int> numbers,
Limits limits)
```

Parameters

`actionType` TestStepActionType

`numbers` List<int>

`limits` Limits

Returns

TestStepAction

createLimitsAction(TestStepActionType, List<string>, Limits)

Creates a TestStepAction with limits using step names.

```
public TestStepAction createLimitsAction(TestStepActionType actionType, List<string> names,
Limits limits)
```

Parameters

`actionType` TestStepActionType

`names` List<string>

`limits` Limits

Returns

TestStepAction

createTestStepAction(TestStepActionType, List<int>)

Creates a TestStepAction based on test step numbers.

```
public TestStepAction createTestStepAction(TestStepActionType actionType, List<int> numbers)
```

Parameters

`actionType` TestStepActionType

`numbers` List<int>

Returns

TestStepAction

createTestStepAction(TestStepActionType, List<string>)

Creates a TestStepAction based on test step names.

```
public TestStepAction createTestStepAction(TestStepActionType actionType,
List<string> names)
```

Parameters

`actionType` TestStepActionType

`names` List<string>

Returns

TestStepAction

toJSON()

Converts the AdaptiveRequest to a formatted JSON string.

```
public string toJSON()
```

Returns

string

A JSON string, or "" in case of serialization error.

Class AdaptiveCommunication

Namespace: Teradyne.FA.DIA.DAS.Adaptive

Assembly: Adaptive.dll

This class manages RabbitMQ communication to send adaptive request to the tester host.

```
public class AdaptiveCommunication
```

Inheritance

object ← AdaptiveCommunication

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdaptiveCommunication(string)

```
public AdaptiveCommunication(string serverIP)
```

Parameters

`serverIP` string

Fields

RMQ_CLIENTNAME

```
public const string RMQ_CLIENTNAME = "AMP DAS RabbitMQ Sender"
```

Field Value

string

RMQ_EXCHANGE

```
public const string RMQ_EXCHANGE = "AMP_DAS_Exchange"
```

Field Value

[string](#)

RMQ_PORT

```
public const int RMQ_PORT = 5672
```

Field Value

[int](#)

RMQ_PWD

```
public const string RMQ_PWD = "TER-ADAPTIVE-P4SSW0RD"
```

Field Value

[string](#)

RMQ_QUEUE

```
public const string RMQ_QUEUE = "AMP_DAS_Queue"
```

Field Value

[string](#)

RMQ_ROUTING

```
public const string RMQ_ROUTING = "AMP_DAS_RoutingKey"
```

Field Value

string

RMQ_SCHEME

```
public const string RMQ_SCHEME = "amqp://"
```

Field Value

string

RMQ_USER

```
public const string RMQ_USER = "AMP-adaptive"
```

Field Value

string

RMQ_VHOST

```
public const string RMQ_VHOST = "TER-AMP-vhost"
```

Field Value

string

connectionFactory

```
protected ConnectionFactory connectionFactory
```

Field Value

ConnectionFactory

m_basicProperties

```
protected IBasicProperties m_basicProperties
```

Field Value

IBasicProperties

m_channel

```
protected IModel m_channel
```

Field Value

IModel

m_connection

```
protected IConnection m_connection
```

Field Value

IConnection

m_consumer

```
protected EventingBasicConsumer m_consumer
```

Field Value

EventingBasicConsumer

m_replyQueueName

```
protected string m_replyQueueName
```

Field Value

string

m_respQueue

```
protected BlockingCollection<string> m_respQueue
```

Field Value

BlockingCollection<string>

rmqServerIP

```
protected string rmqServerIP
```

Field Value

string

Methods

Close()

Close the connection to the RabbitMQ server if it is open.

```
public void Close()
```

Remarks

Attempting to call SendAction(AdaptiveControlCommandBase)
SendAction(AdaptiveControlCommandBase) after closure will result in undefined behavior.

InitBasicProperties()

```
protected void InitBasicProperties()
```

InitConnection()

```
protected bool InitConnection()
```

Returns

[bool](#)

InitConsumerReceived()

```
protected void InitConsumerReceived()
```

Open()

Open a connection to the RabbitMQ server.

```
public bool Open()
```

Returns

bool

SendAction(string)

Send an Adaptive Action to the AMP server.

```
public string SendAction(string message)
```

Parameters

`message` string

Adaptive Action to send. Must be in properly-formatted JSON to be processed correctly.

Returns

string

Response from AMP as a string.

Remarks

See documentation for formatting details. Will throw an InvalidOperationException if the connection is not open.

SendAction(AdaptiveCommand)

Send an Adaptive Action to the AMP server.

```
public string SendAction(AdaptiveCommand command)
```

Parameters

`command` AdaptiveCommand

Returns

string

Response from AMP as a string.

Remarks

See documentation for formatting details. Will throw an InvalidOperationException if the connection is not open.

SendActionAsync(string)

Asynchronously send an Adaptive Action to the AMP server.

```
public Task<string> SendActionAsync(string message)
```

Parameters

`message` string

Adaptive Action to send. Must be in properly-formatted JSON to be processed correctly.

Returns

Task<string>

Response from AMP as a string.

Remarks

See documentation for formatting details. Will throw an InvalidOperationException if the connection is not open.

SendActionAsync(AdaptiveCommand)

Asynchronously send an Adaptive Action to the AMP server.

```
public Task<string> SendActionAsync(AdaptiveCommand command)
```

Parameters

`command` AdaptiveCommand

Returns

Task <string>

Response from AMP as a string.

Remarks

See documentation for formatting details. Will throw an InvalidOperationException if the connection is not open.

Class AdaptiveRequest

Namespace: Teradyne.FA.DIA.DAS.Adaptive

Assembly: Adaptive.dll

Represents an adaptive request sent to the ATE system, including test step actions, enable word changes, and target information.

```
public class AdaptiveRequest
```

Inheritance

object ← AdaptiveRequest

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdaptiveRequest()

Initializes a new instance of the AdaptiveRequest class and sets the timestamp to the current UTC time.

```
public AdaptiveRequest()
```

Properties

DAS_IDENTIFIER

The identifier of the DAS instance generating the request.

```
[JsonRequired]
public string DAS_IDENTIFIER { get; set; }
```

Property Value

string

ENABLEWORD_ACTIONS

A list of enable word actions to perform (e.g., ENABLE, DISABLE).

```
public List<EnableWordAction> ENABLEWORD_ACTIONS { get; set; }
```

Property Value

List<EnableWordAction>

TARGET

The target system or software intended to receive the request.

```
[JsonRequired]
public AdaptiveRequestTarget TARGET { get; set; }
```

Property Value

AdaptiveRequestTarget

TESTSTEP_ACTIONS

A list of test step actions to perform (e.g., ENABLE, DISABLE, UPDATE_LIMITS).

```
public List<TestStepAction> TESTSTEP_ACTIONS { get; set; }
```

Property Value

List<TestStepAction>

TIME_STAMP

The timestamp when the request was created.

```
[JsonRequired]
public DateTime TIME_STAMP { get; set; }
```

Property Value

DateTime

Enum AdaptiveRequestTarget

Namespace: Teradyne.FA.DIA.DAS.Adaptive

Assembly: Adaptive.dll

Enum specifying the target component for an adaptive request.

```
[JsonConverter(typeof(StringEnumConverter))]
public enum AdaptiveRequestTarget
```

Fields

`ATE_SOFTWARE = 0`

Indicates that the adaptive request is intended for the ATE software (test executive).

Class EnableWordAction

Namespace: Teradyne.FA.DIA.DAS.Adaptive

Assembly: Adaptive.dll

Represents an action applied to a set of enable words for adaptive behavior.

```
public class EnableWordAction
```

Inheritance

object ← EnableWordAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ACTION

The action to perform on the enable words (ENABLE or DISABLE).

```
[JsonRequired]
public EnableWordActionType? ACTION { get; set; }
```

Property Value

EnableWordActionType?

ENABLEWORDS

A list of enable word identifiers to be affected by the action.

```
[JsonRequired]
public List<string> ENABLEWORDS { get; set; }
```

Property Value

List<string>

Enum EnableWordActionType

Namespace: Teradyne.FA.DIA.DAS.Adaptive

Assembly: Adaptive.dll

Specifies the type of action to apply on enable words in adaptive control.

```
[JsonConverter(typeof(StringEnumConverter))]
public enum EnableWordActionType
```

Fields

`DISABLE = 1`

Disables a feature or test based on the enable word.

`ENABLE = 0`

Enables a feature or test based on the enable word.

Class Limits

Namespace: Teradyne.FA.DIA.DAS.Adaptive

Assembly: Adaptive.dll

Defines numerical limits and metadata associated with a test step or limit configuration.

```
public class Limits
```

Inheritance

object ← Limits

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

HI_LIMIT

Upper limit value (can be null).

```
[JsonRequired]
public double? HI_LIMIT { get; set; }
```

Property Value

double?

HLM_SCALE

Scale factor applied to the upper limit (optional).

```
public int? HLM_SCALE { get; set; }
```

Property Value

int?

LIMIT_NAMES

Names of the limits involved (e.g., measurement names or parameter labels).

```
public List<string> LIMIT_NAMES { get; set; }
```

Property Value

List<string>

LIMIT_NUMBERS

Numeric IDs corresponding to the limits.

```
public List<int> LIMIT_NUMBERS { get; set; }
```

Property Value

List<int>

LLM_SCALE

Scale factor applied to the lower limit (optional).

```
public int? LLM_SCALE { get; set; }
```

Property Value

int?

LO_LIMIT

Lower limit value (can be null).

```
[JsonRequired]
public double? LO_LIMIT { get; set; }
```

Property Value

double?

Class TestStepAction

Namespace: Teradyne.FA.DIA.DAS.Adaptive

Assembly: Adaptive.dll

Represents an action that can be applied to one or more test steps in an adaptive test scenario.

```
public class TestStepAction
```

Inheritance

object ← TestStepAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ACTION

The type of action to apply (e.g., ENABLE, DISABLE).

```
[JsonRequired]
public TestStepActionType ACTION { get; set; }
```

Property Value

TestStepActionType

LIMITS

The limit configuration to be used if the action is UPDATE_LIMITS.

```
public Limits LIMITS { get; set; }
```

Property Value

Limits

TESTSTEP_NAMES

A list of test step names to which the action applies.

```
public List<string> TESTSTEP_NAMES { get; set; }
```

Property Value

List <string>

TESTSTEP_NUMBERS

A list of test step numbers to which the action applies.

```
public List<int> TESTSTEP_NUMBERS { get; set; }
```

Property Value

List <int>

Enum TestStepActionType

Namespace: Teradyne.FA.DIA.DAS.Adaptive

Assembly: Adaptive.dll

Defines the types of actions that can be taken on test steps in an adaptive scenario.

```
[JsonConverter(typeof(StringEnumConverter))]
public enum TestStepActionType
```

Fields

`CLEAR = 3`

Clears configuration for a test step.

`DISABLE = 1`

Disables a test step.

`ENABLE = 0`

Enables a test step.

`UPDATE_LIMITS = 2`

Updates the limit values for a test step.